EEG Artifact Simulation: Implementation of a selection of artifacts, including eye movements, power line noise, and drifts

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Course: M.Sc. Information Technology, Faculty 5
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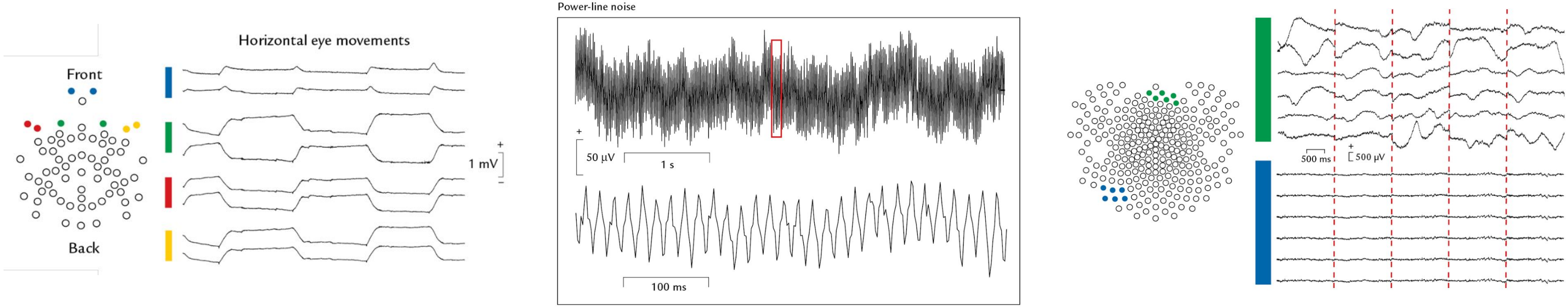
Motivation

- Artifacts - undesirable in EEG recordings



Eye movement, power line noise, and sweating artifacts. Source: Hari, R., & Puce, A. (2017)

- Simulation: improving analysis & artifact removal toolboxes
- Need for open-source, reproducible, flexible artifact simulation

Related Work - Past approaches

- Literature:
  - Extract artifact from data; add to clean(ed) or simulated data
  - Less control over specifics
  - Often: only blink artifacts
  - Often: closed-source
- First steps with Julia: Research Project (03.2025 @ CCS)
  - Two ways to model the eye (CRD, Ensemble)
  - Scalp topography simulation for difference in eye position

References: Klados & Bamidis (2016), Anzolin et al (2021).

Research Questions 🕵️‍♀️

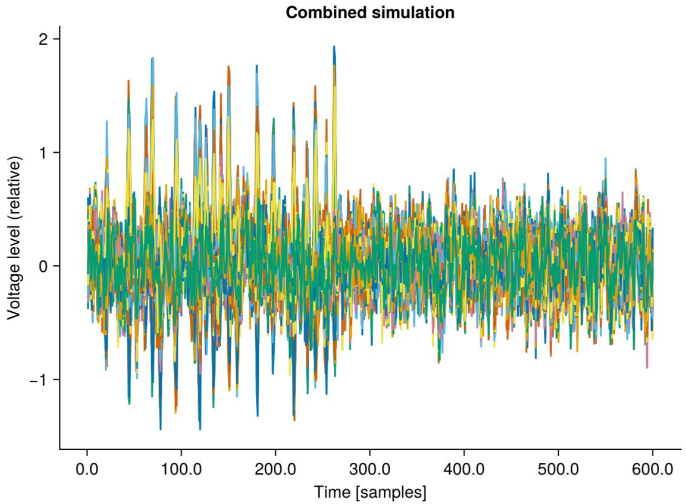
1. What are the characteristics of the measured EEG artifact potentials during an EEG recording?
2. How can we simulate these artifacts?

Goals & Approach

A. Eye movement trajectories: based on Research Project work
B. Power line noise & drifts:
   a. Power line noise: simulate sine waves
   b. Drift: replicate characteristics found from literature review & data analysis
C. Code Interfaces:
   a. UnfoldSim-compatible
   b. Higher-level properties of simulation (e.g. onset, occurrence frequency, …)

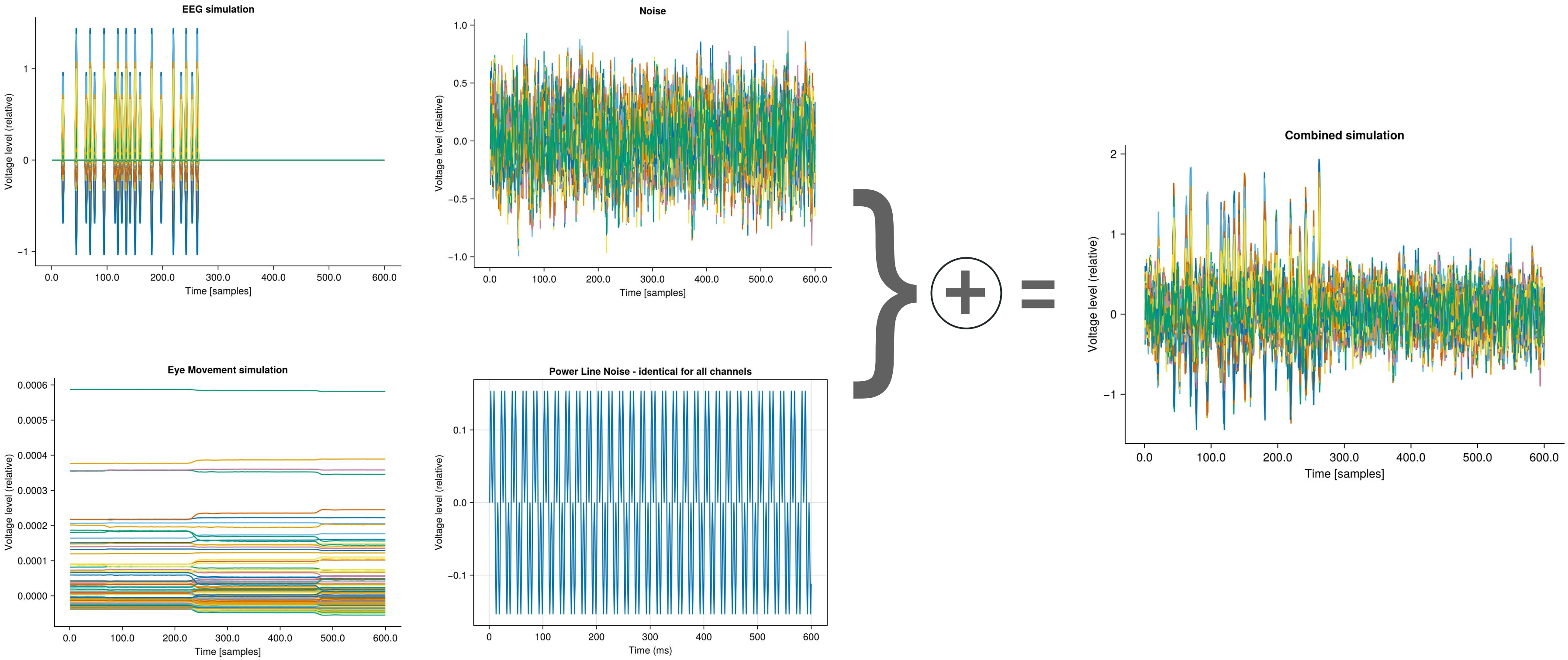
Progress

- Simulation: Eye movement, power line noise (+ EEG, noise)



- Integration with UnfoldSim (in progress)
- Ideas for interfaces and features

Simulation with artifacts



Integration with UnfoldSim

- Concept: “`controlsignal`” for artifact
  - Eye Movement: eye-gaze-direction vectors
  - Power line noise: weights at each channel/time-point

- Modified UnfoldSim function call:

```
simulate(MersenneTwister(1),
design, component, onset,
[noise;
EyeMovement(HREFCoordinates(gaze_trajectory), eyemodel, "ensemble");
PowerLineNoise([], 50, [1 3], [1 0.5], 1000) ]);
```

=> Noise now grouped with Artifacts

Interfaces: controlsignal

- Options for user at different levels of abstraction

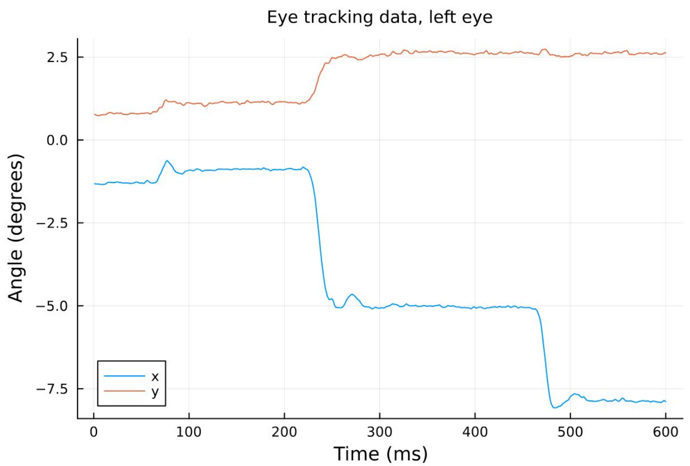
Example: eyeball movement artifact

Higher-level form (user)		
Gaze direction vectors over time	"Virtual screen" gaze coordinates	Fixations + onsets

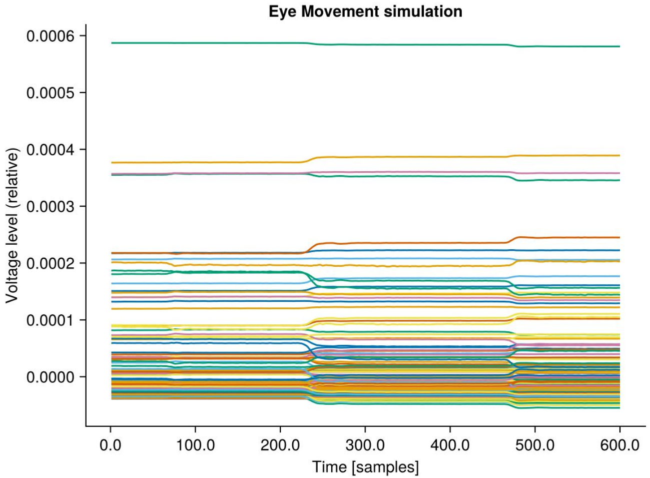
(internal conversion)

Base form (internal use)
Gaze direction vectors over time

Preliminary Results - Eye Movement simulation



Real-data eye movement, left eye gaze x and y traces. Data: Gert et al (2022)



Simulated eye-movement artifact based on the data. (plotted: all channels)

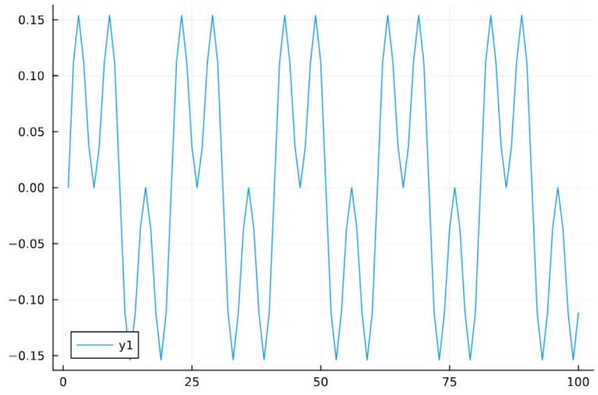
- Challenge: magnitude scaling
  - Levels are different between channels
  - Overall magnitude relative to EEG?

Preliminary Results - Power line noise simulation

- Base frequency, harmonics
- Weighting individual harmonics

- Challenge: magnitude scaling
  - Default weighting of harmonics?
  - Overall magnitude relative to EEG?

```
PowerLineNoise([], 50, [1 3], [1 1], 1000)
# 50Hz, weighted harmonics 1 & 3, sampled @1000Hz
# use default controlsignal
```

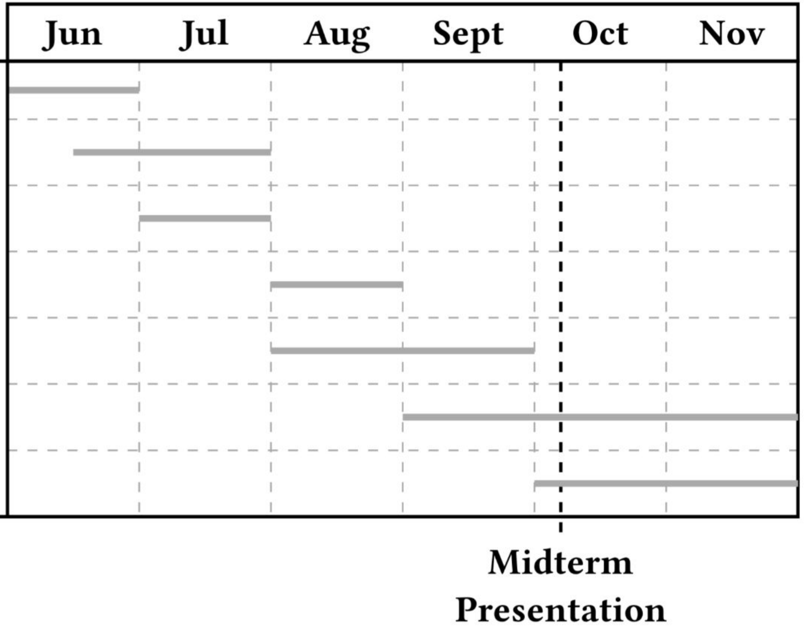


Current challenges + potential solutions

- Magnitude scaling:
  - Sensible defaults, possible options to customise
  - Solution: literature/dataset analysis
- Code interfaces
  - Many possible options
  - Solution: start with basic implementation, build on top
- Drift artifacts
  - Simple first implementation
  - Interfaces + Integrate into UnfoldSim

Schedule: update

Status	Task
🌗	Literature review
✅	Eye movement simulation (Goal A)
➡️	Power line and drift: Dataset analysis (Goal B)
🌗	Power line and drift: Implementation (Goal B)
🌗	Integration with UnfoldSim.jl (Goal C)
➡️	Writing of the thesis
	Buffer / Review / Stretch goals (Goal D-E)



Legend: Status

✅ - Done

🌗 - Partially done

➡️ - Up next

Recap

- EEG simulation packages: brain sources & noise but not (most) artifacts
- Current artifact simulation: non-standard, non-reproducible, closed source
- This thesis:
  - Simulated eye movement artifacts, power line noise
  - Integrate with UnfoldSim.jl (in progress)
  - Next: drift artifact, scaling

Simulate EEG + Artifacts (eye-movement, power line noise, drift) + noise

🤔 Questions?

References

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